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import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

income_df=pd.read_csv(r'C:\Users\HP\Downloads\Inc_Exp_Data.csv')

income_df.head(2)

    Mthly_HH_Income  Mthly_HH_Expense  No_of_Fly_Members  Emi_or_Rent_Amt  \
0                5000                8000                3                2000
1                6000                7000                2                3000

    Annual_HH_Income  Highest_Qualified_Member  No_of_Earning_Members
0                64200                Under-Graduate                1
1                79920                Illiterate                1

income_df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 50 entries, 0 to 49
Data columns (total 7 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Mthly_HH_Income                      50 non-null    int64
1   Mthly_HH_Expense                     50 non-null    int64
2   No_of_Fly_Members                   50 non-null    int64
3   Emi_or_Rent_Amt                     50 non-null    int64
4   Annual_HH_Income                    50 non-null    int64
5   Highest_Qualified_Member             50 non-null    object
6   No_of_Earning_Members                50 non-null    int64
dtypes: int64(6), object(1)
memory usage: 2.9+ KB

income_df.shape

(50, 7)

income_df.describe()

    Mthly_HH_Income  Mthly_HH_Expense  No_of_Fly_Members  Emi_or_Rent_Amt  \
count          50.000000          50.000000          50.000000          50.000000
mean         41558.000000        18818.000000           4.060000        3060.000000
std         26097.908979        12090.216824           1.517382        6241.434948
min           5000.000000          2000.000000           1.000000           0.000000
25%          23550.000000        10000.000000           3.000000           0.000000
50%          35000.000000        15500.000000           4.000000           0.000000
75%          50375.000000        25000.000000           5.000000        3500.000000
max         100000.000000        50000.000000           7.000000       35000.000000

```

	Annual_HH_Income	No_of_Earning_Members
count	5.000000e+01	50.000000
mean	4.900190e+05	1.460000
std	3.201358e+05	0.734291
min	6.420000e+04	1.000000
25%	2.587500e+05	1.000000
50%	4.474200e+05	1.000000
75%	5.947200e+05	2.000000
max	1.404000e+06	4.000000

income_df.describe().T

	count	mean	std	min	25%	\
Mthly_HH_Income	50.0	41558.00	26097.908979	5000.0	23550.0	
Mthly_HH_Expense	50.0	18818.00	12090.216824	2000.0	10000.0	
No_of_Fly_Members	50.0	4.06	1.517382	1.0	3.0	
Emi_or_Rent_Amt	50.0	3060.00	6241.434948	0.0	0.0	
Annual_HH_Income	50.0	490019.04	320135.792123	64200.0	258750.0	
No_of_Earning_Members	50.0	1.46	0.734291	1.0	1.0	

	50%	75%	max
Mthly_HH_Income	35000.0	50375.0	100000.0
Mthly_HH_Expense	15500.0	25000.0	50000.0
No_of_Fly_Members	4.0	5.0	7.0
Emi_or_Rent_Amt	0.0	3500.0	35000.0
Annual_HH_Income	447420.0	594720.0	1404000.0
No_of_Earning_Members	1.0	2.0	4.0

income_df.isna().any()

Mthly_HH_Income	False
Mthly_HH_Expense	False
No_of_Fly_Members	False
Emi_or_Rent_Amt	False
Annual_HH_Income	False
Highest_Qualified_Member	False
No_of_Earning_Members	False

dtype: bool

income_df.columns

```
Index(['Mthly_HH_Income', 'Mthly_HH_Expense', 'No_of_Fly_Members',
      'Emi_or_Rent_Amt', 'Annual_HH_Income', 'Highest_Qualified_Member',
      'No_of_Earning_Members'],
      dtype='object')
```

income_df.columns

```
Index(['Mthly_HH_Income', 'Mthly_HH_Expense', 'No_of_Fly_Members',
```

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        'Emi_or_Rent_Amt', 'Annual_HH_Income', 'Highest_Qualified_Member',
        'No_of_Earning_Members'],
        dtype='object')

income_df["Mthly_HH_Expense"].mean()
np.float64(18818.0)
income_df["Mthly_HH_Expense"].median()
15500.0
print(income_df['Mthly_HH_Expense'].mean())
18818.0
print(income_df['Mthly_HH_Expense'].median())
15500.0
print(income_df['Mthly_HH_Expense'].mode())
0    25000
Name: Mthly_HH_Expense, dtype: int64

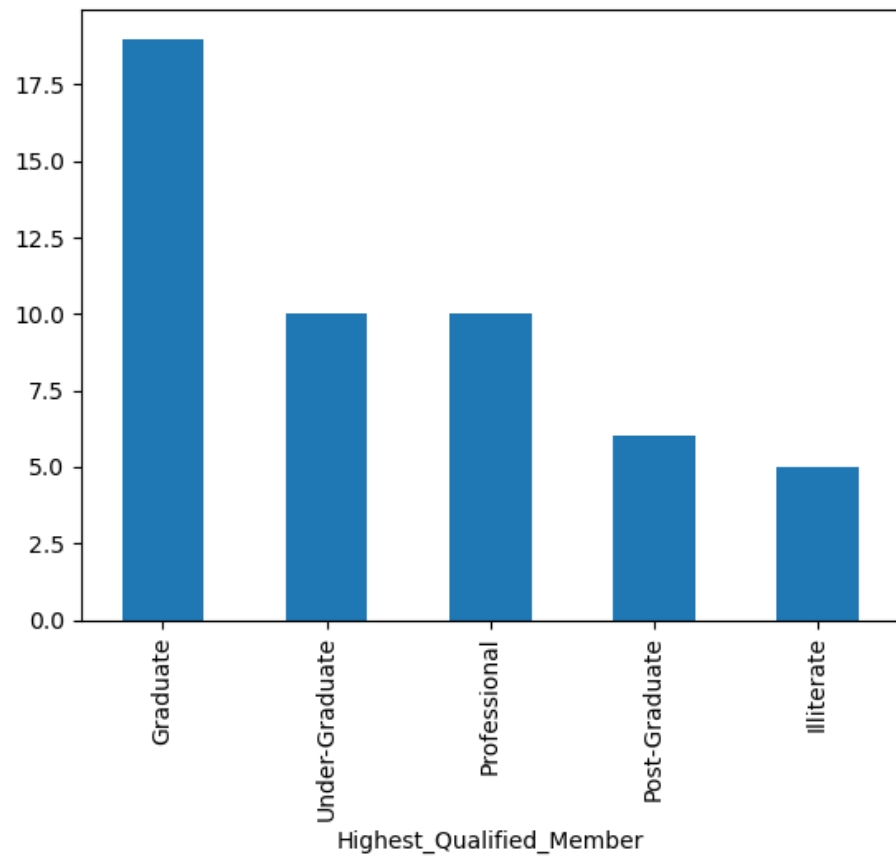
mth_exp_tmp = pd.crosstab(index=income_df["Mthly_HH_Expense"], columns="count")
mth_exp_tmp.reset_index(inplace=True)
mth_exp_tmp[mth_exp_tmp['count'] == income_df.Mthly_HH_Expense.value_counts().max()]

col_0  Mthly_HH_Expense  count
18          25000          8

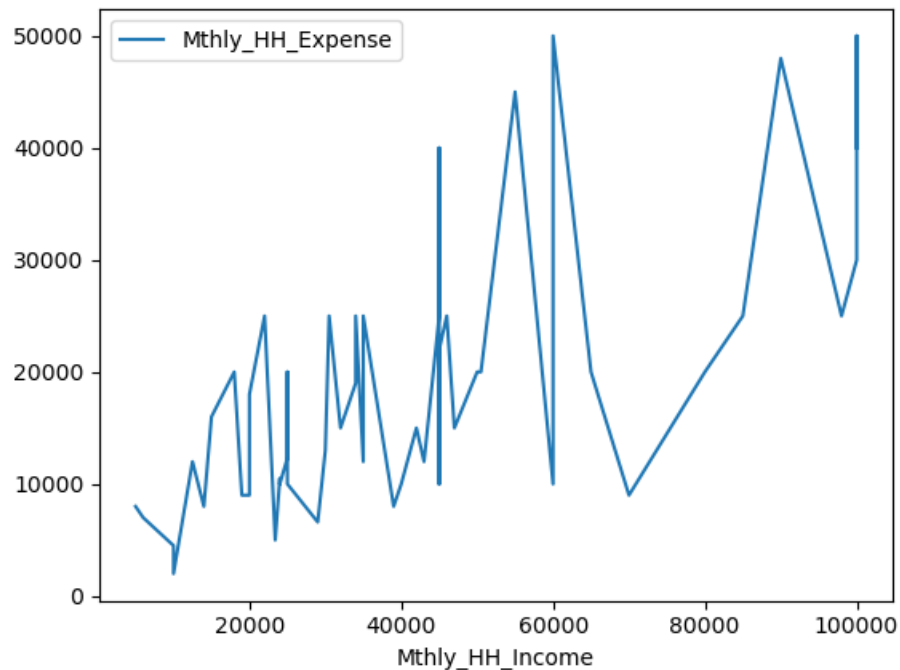
income_df.columns
Index(['Mthly_HH_Income', 'Mthly_HH_Expense', 'No_of_Fly_Members',
       'Emi_or_Rent_Amt', 'Annual_HH_Income', 'Highest_Qualified_Member',
       'No_of_Earning_Members'],
      dtype='object')

income_df["Highest_Qualified_Member"].value_counts().plot(kind="bar")
<Axes: xlabel='Highest_Qualified_Member'>

```



```
income_df.plot(x="Mthly_HH_Income", y="Mthly_HH_Expense")  
<Axes: xlabel='Mthly_HH_Income'>
```



```
pd.DataFrame(income_df.iloc[:,0:5].std().to_frame()).T
      Mthly_HH_Income  Mthly_HH_Expense  No_of_Fly_Members  Emi_or_Rent_Amt  \
0      26097.908979      12090.216824          1.517382      6241.434948

      Annual_HH_Income
0      320135.792123
pd.DataFrame(income_df.iloc[:,0:4].var().to_frame()).T
      Mthly_HH_Income  Mthly_HH_Expense  No_of_Fly_Members  Emi_or_Rent_Amt
0      6.811009e+08      1.461733e+08          2.302449      3.895551e+07
income_df["Highest_Qualified_Member"].value_counts().to_frame().T
Highest_Qualified_Member  Graduate  Under-Graduate  Professional  \
count                      19          10          10

Highest_Qualified_Member  Post-Graduate  Illiterate
count                      6           5
income_df["Highest_Qualified_Member"].value_counts().to_frame().T
Highest_Qualified_Member  Graduate  Under-Graduate  Professional  \
count                      19          10          10

Highest_Qualified_Member  Post-Graduate  Illiterate
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count                                6                5

#Here we need to calculate the coeff of variation

Coeff_of_var_StockA=10/15
print(Coeff_of_var_StockA)
Coeff_of_var_StockB=5/10
print(Coeff_of_var_StockB)

0.6666666666666666
0.5

print(income_df["Mthly_HH_Expense"].quantile(0.75))
25000.0

print(income_df["Mthly_HH_Expense"].quantile(0.50))
15500.0

print(income_df["Mthly_HH_Expense"].quantile(0.25))
10000.0

IQR=income_df["Mthly_HH_Expense"].quantile(0.75)-income_df["Mthly_HH_Expense"].quantile(0.25)
IQR
np.float64(15000.0)

IQR=income_df["Mthly_HH_Expense"].quantile(0.50)-income_df["Mthly_HH_Expense"].quantile(0.25)
IQR
np.float64(5500.0)

income_df['Mthly_HH_Expense'].var()

146173342.85714287

income_df['Mthly_HH_Expense'].std()
12090.216824240286

coef = income_df['Mthly_HH_Expense'].std() / income_df['Mthly_HH_Expense'].mean()
print(coef)
0.6424814977277227

income_df['Annual_HH_Income'].skew
<bound method Series.skew of 0      64200
1      79920
2      112800
3      97200

```

4	147000
5	196560
6	167400
7	216000
8	218880
9	220800
10	278400
11	279840
12	292032
13	316800
14	244800
15	246000
16	261000
17	258000
18	348000
19	385200
20	351360
21	445440
22	330480
23	469200
24	466200
25	449400
26	556920
27	412800
28	488880
29	619200
30	523800
31	507600
32	437400
33	610200
34	596160
35	456840
36	570000
37	581760
38	600600
39	590400
40	590400
41	647400
42	756000
43	1075200
44	1142400
45	885600
46	1152480
47	1404000
48	1032000
49	1320000

```
Name: Annual_HH_Income, dtype: int64>
print(income_df['Annual_HH_Income'].kurtosis())
1.1012908548129197
```